



ALFRED UNIVERSITY

SCHOOL OF ENGINEERING AND APPLIED TECHNOLOGY

Department of Applied Electronics and Embedded Technology

Embedded Systems + CETa Career Transition Program

Course Syllabus & Student Handbook

STUDENT	William A. Green IV
ACADEMIC YEAR	2026-2027 Academic Year
DOCUMENT ID	AU-OAA-ESET301-SYL-2026-02
REVISION	3.0 / v16.3.64
STATUS	Current Governing Document

DISCERE • AEDIFICARE • SERVIRE

Learn. Build. Serve. | Current runtime: v16.3.64 | Core curriculum release: 16.3

CURRENT WEBSITE USE - v16.3.64

Use Classroom for required weekly instruction, Study for reached-material review and repair, Calendar for schedule, Practice for labs/assessments, Progress for completion/sync, and Mastery for retained evidence. Release Notes and build-info.json identify the current runtime.

1. Program Status and Purpose

AU-ESET 301 is a structured personal independent-study program designed to simulate the accountability, sequencing, laboratory evidence, mastery gates, and documentation discipline of a technical course. It is not accredited and awards no academic credit.

AUTHORITY

The accepted core curriculum remains release 16.3. Runtime v16.3.64 governs the live learner experience. Later runtime and content-layer repairs do not silently rewrite the accepted 62-lesson baseline; they supersede specific operational layers through documented release notes.

2. Course at a Glance

Item	Current rule
Course code	AU-ESET 301
Course dates	September 15, 2026 - April 17, 2027
Structure	31 weeks / 62 primary lessons / 24 labs
Primary objective	Build bridge-job-ready electronics, hardware test, validation, and embedded-system evidence while progressing toward an EE / embedded-systems path.
CETa objective	Use the current ETA competency scope plus the Associate CET Study Guide, Sixth Edition. Readiness requires one current Alfred 100-question run at $\geq 85\%$ and one separate current independent ETA/CETa practice assessment at $\geq 85\%$.
Career standards	81 traceability-reconciled Career standards integrated into the 31-week sequence.
Mastery model	Required Classroom stages + evidence-based labs + weekly mastery + cumulative gates. Media consumption alone never establishes mastery.
Cloud Sync	Protocol 2. Local-first records synchronize when configured; the Study Guide module does not block future cloud interaction.

3. Learning Outcomes

- Reason from voltage, current, resistance, power, AC behavior, components, semiconductor devices, power supplies, op-amps, and digital logic at technician / CETa depth.
- Use DMMs, current-limited bench supplies, oscilloscopes, logic analyzers, and SWD/JTAG-style debug tools safely and deliberately.
- Read schematics and datasheets, predict expected values, measure actual behavior, and isolate faults without guesswork.
- Write explainable C / embedded C and Python used for serial communication, data logging, repeatable test automation, and evidence collection.
- Bring up an STM32-class MCU, use GPIO/timers/interrupts/UART, use I2C or SPI, interpret logic-analyzer captures, and document fault handling.
- Demonstrate solder/rework/harness and bench-workmanship evidence appropriate to electronics/test/engineering-technician roles.

- Produce professional engineering evidence: lab notes, calculations, measurement tables, source-controlled code, test plans, failure reports, and portfolio explanations.

4. Required Materials and Learning Sources

Alfred remains the primary teacher. External media is integrated as a deliberate second explanation/reference channel rather than an unbounded parallel curriculum.

Layer	Current role
Classroom Required	247 assigned resources that support first-pass instruction where external video or written literature materially improves comprehension.
Study	81 alternate, reinforcing, or review resources available after material is reached.
Engineering Library	19 professional references such as standards, datasheets, manuals, specifications, application notes, and deeper technical documentation.
CETa Study Guide	64 Required printed pages across 14 weeks; 133 Study/Review pages; 27 Reference/Historical pages. Printed page number -> bundled PDF page uses the verified +9 offset.
Central reading index	214 records: 152 outside-literature sources + 62 Study Guide records. Mixed roles are preserved without creating duplicate obligations.

5. Weekly Classroom Format

- Start Here - prerequisites, purpose, weekly promise, and route orientation.
- CETa Lesson - ground-up instruction, worked reasoning, visual models, guided practice, checks, and semantic evidence.
- Career Lesson - job-facing application and technician/embedded context after prerequisites are taught.
- Teaching Media - Required first-pass sources for the specific sections that benefit from another medium.
- Guided Practice - changed-context work, calculations, troubleshooting, or coding.
- Lab / Application - practical route and required lab knowledge check when that week has a lab.
- Weekly Mastery - cumulative assessment at the weekly target plus any safety-critical lesson checks.

Study has its own independent resume point and a week selector. It does not move Classroom progress backward. Focus Prep appears at the start of sustained Classroom/Study work, not between every lesson section.

6. Completion, Mastery, and Gates

Gate	Current passing evidence
Weekly lab/application	Recorded physical or virtual route plus required lab knowledge-check score $\geq 80\%$ when a lab exists.
Weekly mastery	Weekly mastery score at the course target (normally 80%) plus every safety-critical lesson check required that week.
Week 5 Foundation Gate	Mixed foundation mastery plus independent calculation/build/measurement/fault isolation evidence.
Week 12 Fabrication + cumulative repair	Solder/rework/ESD evidence and repaired cumulative foundation

Gate	Current passing evidence
	gaps.
Week 18 Embedded Interface Core	STM32 + SWD + UART + I2C or SPI + logic analyzer + deliberate fault-injection evidence.
Weeks 21-23 CETa Readiness	One Alfred 100-question run $\geq 85\%$ + one independent current ETA/CETa practice assessment $\geq 85\%$, with weak competencies repaired.
Week 30 Career green light	Common core + Project 1 + role proof + professional application package are explainable and demonstrable.
Week 31 Application Sprint	Apply, interview, log feedback, and convert real market evidence into dated repairs.

7. Assessment and Evaluation Policy

The live site does not require a parallel spreadsheet gradebook or hours log. Completion is evidence- and gate-based. Scores remain useful evidence, but a high average does not override an unsafe or missing practical gate.

- Required lesson checks and semantic tasks prove topic-specific understanding.
- Lab knowledge checks verify route understanding; practical evidence verifies what was actually built/measured.
- Weekly mastery forms use current reviewed evidence and route misses back to the exact teaching area.
- Teaching Media watched/read/reviewed checklists are organizational aids only; they do not create mastery.
- Prior passes are not revoked when Required resources are later restored; Alfred recommends review without rewriting historical completion.

8. Safety, Integrity, and AI Use

- Personal bench work stays low-voltage and current-limited unless a qualified environment explicitly provides the procedures, supervision, and equipment for something else.
- Follow manufacturer and authoritative safety guidance for DMM current measurement, earth-referenced oscilloscope grounds, soldering/rework, ESD control, power supplies, and battery/energy hazards.
- AI may tutor, explain, generate practice, review reasoning, and help debug. It cannot substitute for measurements, mastery, or interview-defensible understanding.
- Portfolio artifacts must represent work you can explain. Document uncertainty and failed attempts instead of polishing away evidence of the engineering process.
- When college/work/family constraints collide, reschedule deliberately. Do not create unhealthy make-up marathons.

9. 31-Week Schedule

Week	Dates	Topic	Lab	Primary close
W01	2026-09-15 to 2026-09-19	Electricity Basics + Safety + Ohm/Watt	LAB-001	Lab + mastery
W02	2026-09-21 to 2026-09-26	DC Networks + KVL/KCL + Voltage Dividers + DMM	LAB-002	Lab + mastery
W03	2026-09-28 to 2026-10-03	DMM + Bench Supply + Oscilloscope	LAB-003	Lab + mastery
W04	2026-10-05 to 2026-10-10	AC Waveforms + R/C/L	LAB-004	Lab + mastery

Week	Dates	Topic	Lab	Primary close
		Behavior		
W05	2026-10-12 to 2026-10-17	Foundation Mastery Check	LAB-005	Lab + mastery
W06	2026-10-19 to 2026-10-24	Reactance + Impedance + Resonance + Filters + Transformers	LAB-006	Lab + mastery
W07	2026-10-26 to 2026-10-31	Components + Magnetism + Motors/Generators + Relays	LAB-007	Lab + mastery
W08	2026-11-02 to 2026-11-07	Semiconductors + Switching Devices	LAB-008	Lab + mastery
W09	2026-11-09 to 2026-11-14	Electronic Power Supplies	LAB-009	Lab + mastery
W10	2026-11-16 to 2026-11-21	Amplifiers + Op-Amps + Gain	LAB-010	Lab + mastery
W11	2026-11-23 to 2026-11-28	Digital Logic + Number Systems	LAB-011	Lab + mastery
W12	2026-11-30 to 2026-12-05	Soldering + Git + Cumulative Repair	LAB-012	Lab + mastery
W13	2026-12-07 to 2026-12-12	Schematics + Troubleshooting	LAB-013	Lab + mastery
W14	2026-12-14 to 2026-12-19	C Programming I + Computer Architecture	LAB-014	Lab + mastery
W15	2026-12-21 to 2026-12-26	C Programming II + Embedded C + State Machines	LAB-015	Lab + mastery
W16	2026-12-28 to 2027-01-02	STM32 Setup + GPIO + SWD Debug	LAB-016	Lab + mastery
W17	2027-01-04 to 2027-01-09	UART + Python Serial	LAB-017	Lab + mastery
W18	2027-01-11 to 2027-01-16	I2C/SPI + Logic Analyzer + Sensors/Actuators	LAB-018	Lab + mastery
W19	2027-01-19 to 2027-01-23	Cabling + Audio/Video + Optical	LAB-019	Lab + mastery
W20	2027-01-26 to 2027-01-30	RF + Telecom + Technician Procedures	LAB-020	Lab + mastery
W21	2027-02-02 to 2027-02-06	CETa Final Review I	LAB-021	Lab + mastery
W22	2027-02-09 to 2027-02-13	CETa Final Review II + Project 1 Design	LAB-022	Lab + mastery
W23	2027-02-16 to 2027-02-20	CETa Readiness Gate + Project Setup	-	Mastery / project
W24	2027-02-23 to 2027-02-27	Project 1 Hardware Bring-Up	-	Mastery / project
W25	2027-03-02 to 2027-03-06	Project 1 Sensor / Interface Integration	-	Mastery / project
W26	2027-03-09 to 2027-03-13	Project 1 Firmware Architecture	-	Mastery / project
W27	2027-03-16 to 2027-03-20	Python Hardware Test +	LAB-023	Lab + mastery

Week	Dates	Topic	Lab	Primary close
		Automated Logger		
W28	2027-03-23 to 2027-03-27	Fault Injection + Troubleshooting + Test Report	-	Mastery / project
W29	2027-03-30 to 2027-04-03	Instrumentation + Solder/Rework/Harness Role Proof	-	Mastery / project
W30	2027-04-06 to 2027-04-10	Portfolio + Resume + Interview Green-Light	LAB-024	Lab + mastery
W31	2027-04-13 to 2027-04-17	Application Sprint + Interview Feedback Loop	-	Mastery / project

10. Document and Version Authority

For detailed original lesson architecture, the v16.3 Curriculum & Instruction Plan remains the frozen baseline. Current Career wording/provenance and prerequisite routing are controlled by the v16.3.42 traceability package. Current Teaching Media architecture is controlled by the v16.3.60 resource layer. Current Study Guide integration is controlled by the v16.3.53-v16.3.61 records. build-info.json and Release Notes identify the current runtime patch.

DOCUMENT CONTROL

When a historical PDF conflicts with the live site, use the current governing document, build-info.json, and the most recent applicable release note. Historical acceptance reports remain evidence of what was verified at their own release; they are not silently rewritten.